

Migrations

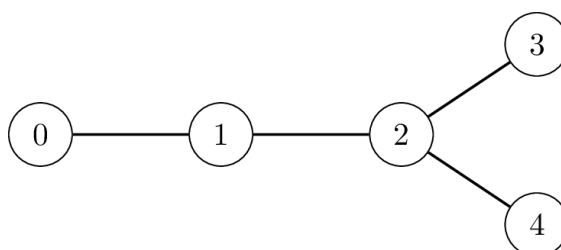
Mize (museum) y' amateka iri kugenzura uko dinazoro (dinosaurs) zagiye zimuka muri Bolivia. Abashakashatsi bagiye babona ibisigazwa byazo muri N site (sites), ziriho ibimenyetso by' imibare guhera kuri 0 kugera kuri $N - 1$ uko imyaka zimaze igenda ingabanuka: site 0 ifite ibisigazwa bya kera cyane, mu gihe site $N - 1$ ifite ibisigazwa bimaze imyaka mike.

Dinazoro zimukiye kuri buri site (uretse site ya 0) ziturutse kuri site runaka imaze igihe kirekire. Kuri buri site i aho $1 \leq i \leq N - 1$, hari site $P[i]$ imaze imyaka myinshi kuyirusha ($P[i] < i$) aho dinazoro zavuye kuri site $P[i]$ zikajya kuri site i .

Site imaze imyaka myinshi runaka ishobora kuba yarimuye dinazoro zijya muri sites zitandukanye zimaze imyaka mike.

Abashakashatsi berekana kwimuka nk' **uregendo rudasubira inyuma** ruri hagati ya sites i na $P[i]$. Menya ko kuri buri sites ebyiri x na y , birashoboka kuva kuri x ujya kuri y unyuze mu ngendo nyinshi zikurikiranye. **Urugendo** ruri hagati ya x na y ni umubare muto ushoboka w' ingendo ziva kuri x zijya kuri y .

Urugero, muri iyi foto hari site $N = 5$, aho $P[1] = 0$, $P[2] = 1$, $P[3] = 2$, kandi $P[4] = 2$. Ushobora kuva kuri 3 ukajya kuri 4 unyuze kuri 2, distance iri hagati yazo ni 2.



Mize irashaka kwerekana itsinda rya sites zifite distance nini ishoboka. Menya ko igisubizo kitagombye kuba cyihariye: nko mu rugero rwerekanwe haruguru, amatsinda $(0, 3)$ na $(0, 4)$ afite distance 3, kandi niyo nini ishoboka. Hano, rimwe muri ariya matsinda rishobora kuba igisubizo.

Mu ntagiriro, imibare ya $P[i]$ **ntago** izwi. Mize yohereje itsinda ry' abashakashatsi gusura sites $1, 2, \dots, N - 1$, gutyo bikurikiranye. Nyuma yo kugera kuri site i ($1 \leq i \leq N - 1$), abashakashatsi bakoze ibi bikurikira:

- Gushakisha $P[i]$, aho zavuye zijya kuri site i .

- Kwanzura niba bagomba kohereza message **itarenze imwe** kuri Mize, bitewe n' amakuru bafite.

Message yoherezwa binyuze kuri satellite ihenze, bishatse kuvuga ko buri message ari umubare uri hagati ya 1 na 20 000, inclusive. Byogeye kandi, abashakashatsi bemerewe kohereza **messages zitarenze** 50 muri rusange.

Akazi kawe ni ugukora uburyo aho:

- Abashakashatsi bahitamo site bagomba kohererezaho message, detse n' umubare wa message.
- Mize ishobora guhitamo sites ebyiri zifite distance nini ishoboka, igendeye kuri messages zavuye kuri buri site, kandi iba izi sites messages zavuyeho.

Kohereza umubare munini unyuze kuri satellite birahenze. Amanota ubona araba agendeye ku mubare munini woherejwe n' ingano ya messages zoherejwe.

Implementation Details

Ugomba gukora function y' itsinda ry' abashakashatsi, n' indi ya Mize.

Function y' **Itsinda ry' abashakashatsi** niyi:

```
int send_message(int N, int i, int Pi)
```

- N : umubare wa sites zifite ibisigazwa.
- i : umubare (index) wa site itsinda rigezeho.
- P_i : umubare wa $P[i]$.
- Iyi function ihamagarwa inshuro $N - 1$ kuri buri test case, bitondekanyijwe muri ubu buryo $i = 1, 2, \dots, N - 1$.

Iyi function igomba kureturninga $S[i]$ yerekana igikorwa abashakashatsi bakoze bari kuri site i :

- $S[i] = 0$: abashakashatsi bahitamo kutagira message n' imwe bohereza bari kuri site i .
- $1 \leq S[i] \leq 20\,000$: abashakashatsi bohereza message ifite umubare $S[i]$ bari kuri site i .

Function ya **Mize**:

```
std::pair<int,int> longest_path(std::vector<int> S)
```

- S : array ifite length N aho:
 - $S[0] = 0$.
 - Kuri buri $1 \leq i \leq N - 1$, $S[i]$ ni umubare uba returned na `send_message(N, i, Pi)`.
- Iyi function ihamagarwa inshuro imwe kuri buri test case.

Iyi function igomba kureturninga itsinda rya sites (U, V) zifite distance nini ishoboka hagati yazo.

Mu gutanga amanota, Program ihamagara iyi function **inshuro ebyiri**.

- Ku nshuro yambere umaze guhamagara program:
 - `send_message` ihamagarwa inshuro $N - 1$.
 - **Program yawe ishobora kubika amakuru uko igenda ihamagarwa.**
 - Ibyareturninzwe birabikwa na system ikosora.
 - Hari cases aho grader iba **adaptive**. Bishatse kuvuga ko umubare wa $P[i]$ muri `send_message` ugendanye n' ibyo abashakashatsi bakoze muri functions zabanje.
- Ku nshuro ya kabiri yo guhamagara program:
 - `longest_path` ihamagarwa inshuro imwe, values zibikwa na grading system ku nshuro yambere.

Constraints

- $N = 10\,000$
- $0 \leq P[i] < i$ kuri buri i aho $1 \leq i \leq N - 1$.

Subtasks and Scoring

Subtask	Score	Additional Constraints
1	30	Site 0 n' iyindi site runaka zifite distance nini ishoboka.
2	70	Ntazindi constraints.

Reka Z ibe umubare munini ushoboka muri array S , kandi M ibe umubare wa messages zoherezwa n' abashakashatsi.

Kuri test case runaka, iyo hari imwe muri condition zigiye gutangwa ikunda, solution yawe irabona amanota 0 (urabona `Output isn't correct` muri CMS):

- Hari nibura umubare umwe muri S utari wo.
- $Z > 20\,000$ cyangwa $M > 50$.
- Function `longest_path` itanze result itari yo.

Bitabaye ibyo, amanota ya subtask 1 abonwa gutya:

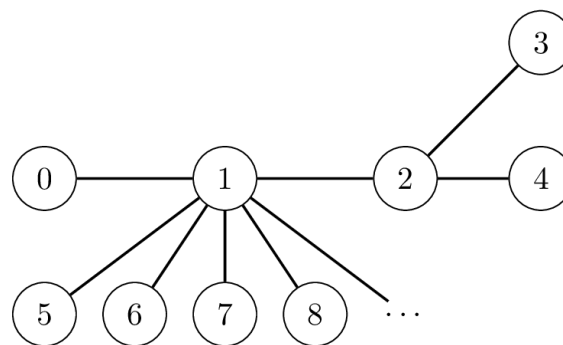
Condition	Score
$9\,998 \leq Z \leq 20\,000$	10
$102 \leq Z \leq 9997$	16
$5 \leq Z \leq 101$	23
$Z \leq 4$	30

Amanota ya subtask 2 abonwa gutya:

Condition	Score
$5 \leq Z \leq 20\,000$	$35 - 25 \log_{4000} \left(\frac{Z}{5} \right)$
$Z \leq 4$ kandi $32 \leq M \leq 50$	40
$Z \leq 4$ and $8 < M < 32$	$70 - 30 \log_4 \left(\frac{M}{8} \right)$
$Z \leq 4$ and $M \leq 8$	70

Example

Reka $N = 10\,000$. Tekereza igihe $P[1] = 0$, $P[2] = 1$, $P[3] = 2$, $P[4] = 2$, and $P[i] = 1$ for $i > 4$.



Tekereza ko strategy y' abashakashatsi ari ukohereza message $10 \cdot V + U$ mu gihe itsinda rya sites (U, V) rifite distance nini ishoboka rihindutse kubera `send_message` call.

Mbere, itsinda rifite distance nini ni $(U, V) = (0, 0)$. Tekereza ko program ihamagaye muri ubu buryo ku nshuro yambere:

Procedure call	(U, V)	Return value
<code>send_message(10000, 1, 0)</code>	$(0, 1)$	10
<code>send_message(10000, 2, 1)</code>	$(0, 2)$	20
<code>send_message(10000, 3, 2)</code>	$(0, 3)$	30
<code>send_message(10000, 4, 2)</code>	$(0, 3)$	0

Menya ko muri calls zose zisigaye value ya $P[i]$ ari 1. Bishatse kuvuga ko itsinda rifite distance nini ishoboka ridahinduka, kandi ko abashakashatsi batohereza izindi messages.

Noneho ku nshuro ya kabiri program ihamagara muri ubu buryo:

```
longest_path([0, 10, 20, 30, 0, ...])
```

Mize isoma message yanyuma yoherejwe n' abashakashatsi, ariyo $S[3] = 30$, hanyuma ikabona ko $(0, 3)$ ari itsinda rya site rifite distance nini ishoboka. Bishatse kuvuga ko, iyi function ireturninga $(0, 3)$.

Menya ko ubu buryo budatuma buri gihe Mize ishobora kumenya itsinda rifite distance nini ishoboka.

Sample Grader

The sample grader ihamagara `send_message` na `longest_path` icyarimwe, bitandukanye na grader ya nyayo.

Input format:

```
N
P[1] P[2] ... P[N-1]
```

Output format:

```
S[1] S[2] ... S[N-1]
U V
```

Menya ko ushobora gukoresha sample grader wifashishije umubare ushatse wa N .